Fortifying Centers and Edges: Multidomain Feature Learning Meets Hyperspectral Image Classification

Hao Shi, Youqiang Zhang, *Member, IEEE*, Guo Cao, *Member, IEEE*, and Di Yang

***Abstract*— Hyperspectral image (HSI) classification is an important topic in remote sensing tasks, aiming to exploit the spectral and spatial information in HSI to assign category labels to each pixel. Recently, numerous deep learning (DL) methods, such as convolutional neural network (CNN) and transformer, have been introduced and achieved promising results. These DL-based models tend to use patch-based input patterns, where a patch is extracted around a central pixel and the neighboring pixels serve as auxiliary roles. However, compared to developing novel models, the characteristics of the patch-based HSI classification framework have not been adequately explored. Most existing patch-based DL methods tend to overlook the potential relationships between the central pixel and its neighboring pixels. In addition, the unique spectral characteristics of HSI (e.g., adjacent bands possess higher correlation) require special attention. Moreover, the design of feature extractors for HSIs appears to be constrained primarily to the spatial and spectral domains. To cope with the above problems, we propose an HSI classification specifically designed model (HSIC-SDM), which encompasses three branches—frequency feature learning, spatial feature learning, and spectral feature learning. The frequency feature learning branch employs a Fourier-based feature extractor to acquire representations in the frequency domain. The spatial feature learning branch emphasizes the significance of central pixel and edge information. The spectral feature learning branch concerns the high correlations in the adjacent spectral bands. Multidomain features derived from three branches provide diverse perspectives for the final classification. Extensive experiments on three datasets demonstrate the superiority of the proposed method over the state-of-the-art (SOTA) compared methods. The code of our work will be available publicly at https://github.com/Tikiten/HSIC-SDM.**

***Index Terms*— Convolutional neural network (CNN), edge detection, feature fusion, Fourier transform, hyperspectral image (HSI) classification, transformer.**

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I. Introduction

HYPERSPECTRAL images (HSIs) generally involve hundreds of narrow spectral channels, which encompass continuous spectral signatures and exhibit different spatial morphological features. HSI is an important link in numerous remote sensing applications, including but not limited to classification [1], [2], [3], environmental management [4], and precision agriculture [5]. HSI classification, as a pivotal aspect of hyperspectral data analysis, aims at assigning each pixel to a specific label or class based on the discriminative features of various materials. This task holds significant importance across a wide range of practical applications.

Compared to the conventional RGB image classification, HSI classification is more arduous due to the curse of dimensionality [6]. In the early stages, numerous machine learning algorithms have been applied to HSI categorization. Specifically, many spectral-based methods were developed to explore rich spectral features for HSI classification, including support vector machine (SVM) [7] and principal component analysis (PCA) [8]. Moreover, there are also methods that consider the joint spectral–spatial characteristics of HSI and integrate spatial information to improve classification results [9], [10]. However, traditional methods are constrained by shallow feature extraction and cumbersome manual design, demonstrating insufficient learning ability and adaptability in addressing various HSI scenarios.

Recently, deep learning (DL) techniques have flourished in various computer vision fields. Consequently, DL-based hyperspectral classification research has gained substantial momentum, yielding numerous successful cases. Among them, convolutional neural network (CNN), being one of the most widely adopted models, has garnered plenty of attention in HSI classification. In general, CNN-based HSI classification models involve the extraction of multiple HSI patches from the original HSI data, which are fed into the network to determine a category label for the central pixel of each patch. More recently, numerous CNN-based HSI classification approaches have been proposed, ranging from 1D-CNN to 3D-CNN [11], [12], [13], and from single networks to hybrid networks. A wide range of CNN variants have been developed to obtain better classification results [14], [15], [16]. Given the substantial impact of convolutional kernels on feature learning, some researchers have endeavored to design different kernels. For example, recent studies have employed dilated

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convolution and irregular convolution techniques [17], [18]. Besides, two-branch or three-branch networks [19], [20] have gained considerable attention in recent years. These networks excel at extracting HSI features from multiple perspectives to achieve richer representations, which is beneficial to promote classification results. Nevertheless, CNN-based methods do have some drawbacks. First, owing to the constraints imposed by the convolutional kernel, the receptive field of CNN-based methods is restricted to a limited square region, which hinders the network from learning long-distance dependencies [21]. Second, traditional CNN-based methods have difficulty in modeling land covers with intricate structures [22].

To assist CNNs in better feature learning, many techniques have been developed, with the self-attention (SA) mechanism standing out as the most prominent [23], [24], [25]. Recently, there has been a lot of work that extends beyond the view of the SA mechanism as an auxiliary to CNNs. Instead, researchers place greater emphasis on the role of SA mechanism when designing their networks and, in some cases, construct networks entirely centered around the SA mechanism. Consequently, a series of approaches, referred to as transformer-based methods, have been developed [26], [27], [28], [29], [30].

Moreover, a recent study has proposed that the success of transformer architecture can primarily be attributed to the general architecture of the transformer encoder and the core component multihead SA (MHSA) is not mandatory [31]. With this concept in mind, researchers have designed different spatial multilayer perceptrons (MLPs) to supplant MHSA as token mixers and achieved positive performance [32], [33]. Besides, there are other explorations of token mixers such as those utilizing Fourier transform [34], [35], [36], [37]. These efforts inspire us to go beyond the limitations of CNNs or transformers to design efficient feature extractors.

Based on the above analysis, it can be observed that many existing works tend to transplant network structures directly from computer vision tasks targeting RGB images to HSI classification. This leads to the oversight of distinctive attributes inherent in HSIs and the specific characteristics of HSI classification. We attribute the deficiencies in existing works to the following problems.

Problem 1: Many HSI classification models define a dominant evolutionary path, transitioning from CNN-based methods to SA-enhanced CNN-based methods, and ultimately embracing transformer-based methods. This path deserves to be explored further, i.e., new feature extractors deserve to be investigated. Given that both CNNs and transformers learn features in the spatial domain, it can be considered to design a feature extractor in a different domain such as the frequency domain.

Problem 2: In terms of HSIs, HSIs distinguish themselves from RGB images by virtue of their abundant spectral bands, which offer a wealth of spectral signatures. Moreover, it is noteworthy that adjacent spectral bands exhibit high correlation, while long-range spectral bands display lower correlation. Many studies fail to pay attention to this point. For instance, all spectral bands might be processed at one time, resulting in the loss of some important distinguishing information.

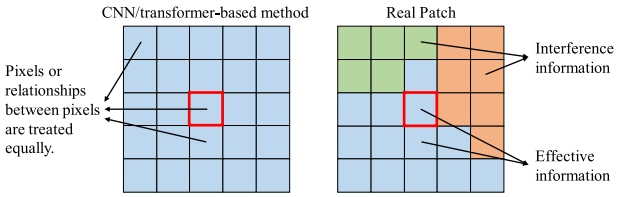


Fig. 1. CNN-based or transformer-based methods commonly ignore the real condition within a patch (take the patch size of 5 × 5 for example and the pixel in the red box is the central pixel).

Problem 3: In terms of HSI classification frameworks, they can be broadly categorized into two groups: patch-based framework and patch-free framework. In our research and comparative analysis, we focus on patch-based methods. Patch-based methods crop patches centered on each pixel in the HSI for network training and testing. According to this data preprocessing approach, the central pixel in a patch should be particularly valued rather than being treated equally with other pixels [38]. However, CNN-based approaches consider pixels in an HSI patch with the same weight. In the transformer-based approaches, for the query pixels, whether neighboring pixels or long-range pixels, the relationships between them are treated equally and uniformly generated by the transformer encoder. Fig. 1 depicts the paradox. In addition, pixels located at the edges of different category regions are more susceptible to being misclassified in patch-based methods due to their heightened susceptibility to interference from surrounding pixels belonging to other categories.

Based on the foregoing analysis, we propose an HSI classification specifically designed model (HSIC-SDM) to efficiently accomplish classification by targeting the distinctive attributes of HSIs and the specific characteristics of patch-based HSI classification. The proposed model can alleviate the three abovementioned problems in HSI classification. Concretely, HSIC-SDM comprises three primary components: frequency feature learning, spatial feature learning, and spectral feature learning. They make it possible to learn features from HSIs in different domains, providing diverse perspectives for HSI classification. The main contributions can be summarized as follows.

1) Three problems are concluded for patch-based HSI classification. They provide a comprehensive and novel perspective when designing new models.
2) According to Problem 1, three branches are constructed to extract multidomain features. The Fourier-transform-based frequency modulator is our efficient feature extractor, which is used in two branches. It offers a new perspective for feature extraction in HSI classification.
3) Considering Problem 2, we adaptively group highly correlated neighboring bands and use a spectral frequency modulator to perform spectral feature learning in each group. Compared to processing all spectral bands at one time, this design is more in line with the characteristics of HSIs since the discriminative information of certain materials is usually contained in one or a few spectral ranges (e.g., green plants are sensitive to the red and near-infrared bands).
4) To tackle Problem 3, against the characteristics of patch-based HSI classification, we propose the

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center-edge-enhanced module to emphasize the value of central pixel and edge information. This new feature extractor is also a solution to Problem 1. In the module, the original central pixel vector is adopted as a query vector, imitating the operation of SA to consistently reinforce the importance of the center position. Besides, we introduce the idea of edge detection in image processing and employ the Sobel [39] operator to enhance edge information, aiming to reduce misclassification of pixels located at the edges.

The remainder of this article is organized as follows. Section II elaborates on the proposed HSIC-SDM. Section III provides the experimental results as well as the related analysis. Section IV concludes this article and discusses potential future work.

II. Related Work

A. CNN-Based Methods and Attention-Aided CNNs

Research on CNNs for HSI classification has been a hot spot. Li et al. [11] proposed an HSI classification model based on a pixel-level CNN framework to fully leverage the advantages of CNN. Roy et al. [12] proposed a hybrid spectral CNN (HybridSN) for HSI classification, which combines 2D-CNN and 3D-CNN to facilitate the joint spatial–spectral feature representation. Li et al. [13] designed a light 3D-CNN HSI classification framework. In addition, Song et al. [14] introduced residual structures [40] to form a deep residual network, aiming to merge features from various layers. Zhong et al. [15] proposed an end-to-end spectral–spatial residual network (SSRN). Zhang et al. [16] incorporated a multiscale analysis to construct a multiscale CNN for classification. Besides, different forms of convolution, such as dilated convolution [17] and irregular convolution [18], have been used for HSI classification, achieving positive results. Shi et al. [19] and Tu et al. [20] designed CNNs with multiple branches to enrich the generated features.

To acquire more accurate representations, the attention mechanism is often employed to aid in CNN-based HSI classification. In [23], the attention-based adaptive spectral–spatial kernel improved residual network (A2S2K) incorporated the attention mechanism within the 3D-CNN residual block, effectively enhancing feature extraction. In addition, the residual spectral–spatial attention network (RSSAN) [24] integrated spectral–spatial attention modules into a network with residual connections, enabling adaptive learning of spectral and spatial features. Shi et al. [25] devised a multibranch hybrid CNN that incorporates the attention mechanism and operates at multiple resolutions.

B. Transformer-Based Methods and Other Token Mixers

Recently, applying transformers to HSI classification has also been a popular area of research. Hong et al. [26] treated spectral signatures as sequential data and transformed preprocessed local spectral features into spectral tokens. These tokens were subsequently input into the transformer encoder, incorporating cross-layer skip connections to facilitate feature learning. Yu et al. [27] employed a pure transformer encoder to facilitate feature learning from serialized HSIs, showcasing the applicability of the transformer structures in HSI classification. Similarly, Yang et al. [28] conducted global and local feature extraction on HSIs with 3-D convolutions, which generates an improved feature sequence for the transformer encoder. Furthermore, Sun et al. [29] utilized hybrid CNNs to extract shallow spectral–spatial features prior to the transformer encoder. Song et al. [30] presented the bottleneck spatial–spectral transformer (BS2T), which integrates 3-D CNNs with specially designed MHS2A modules. It is worth noting that certain studies mentioned earlier also incorporated convolutional layers within their networks and can therefore be labeled as CNN–transformer methods.

The core element of the transformer is the attention module, which serves as a token mixer. Recently, various other forms of token mixers have been gaining attention. Many studies have adopted MLPs as token mixers and proposed numerous MLP-based models, e.g., gMLP [32] and ResMLP [33]. Moreover, token mixers based on Fourier transform also exhibit good performance. Rao et al. [34] presented the global filter network (GFNet), which utilizes the Fourier transform and inverse Fourier transform for efficient feature modeling. Similarly, Guibas et al. [35] proposed an adaptive Fourier neural operator (AFNO) as a token mixer that learns representations in the frequency domain. Furthermore, Shi et al. [36], [37] proposed F3Net and MHCFormer, both effectively integrating Fourier transform with CNN or transformer architectures and yielding satisfactory results in HSI classification.

C. Factor Analysis

Factor analysis (FA) is a statistical method for describing variability among observed related variables in terms of a potentially small number of unobserved variables named factors. Observed variables are modeled as linear combinations of latent factors plus error terms, and thus, FA can be considered as a special case of errors-in-variables models. Simply put, FA is a technique that reduces a large number of variables into fewer numbers of factors. It extracts the maximum common variance from all variables and puts them into a common score, which can be used for further analysis.

Assume that there are a number of s observations with a set of t common factors $f_{i,j}$ ($t < s$). Each individual has t of their own common factors. These common factors are related to the observations using the factor loading matrix ($L \in \mathbb{R}^{s\times t}$), which for a single observation is calculated as

$$x_{i,p} - \mu_i = l_{i,1} f_{1,p} + \cdots + l_{i,t} f_{t,p} + \varepsilon_{i,p} \tag{1}$$

where $x_{i,p}$ stands for the value of the ith observation of the pth individual, μ_i represents the observation mean for the ith observation, $l_{i,j}$ is the loading for the ith observation of the jth factor, $f_{j,p}$ denotes the value of the jth factor of the pth individual, and $\varepsilon_{i,p}$ is the (i, p)th unobserved stochastic error term with zero mean and finite variance.

III. Methodology

A. Overall Architecture

As shown in Fig. 2, the proposed HSIC-SDM is a multibranch patch-based framework. It is worth noting that the

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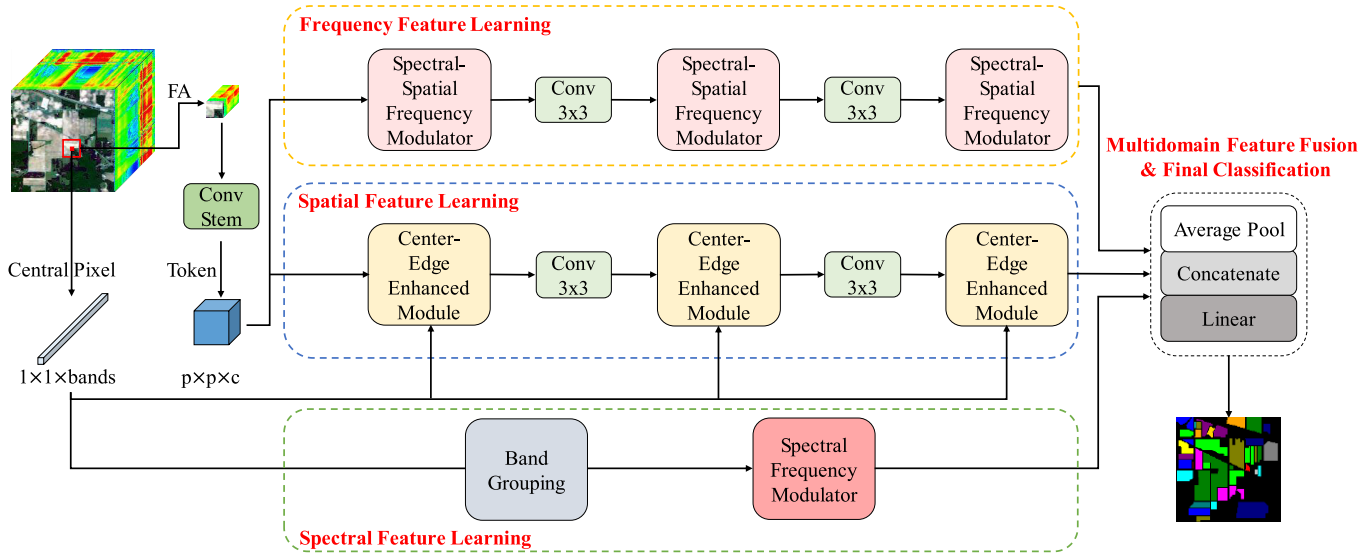


Fig. 2. Overall architecture of the proposed HSIC-SDM, which mainly consists of three branches. Three branches provide features from the frequency, spatial, and spectral domains. Conv3 × 3 means a convolution layer with size of 3 × 3, aiming to downsample feature maps for hierarchical feature learning. The inputs to the three branches are not identical. Multidomain features from different branches are eventually fused in the way of concatenation for classification.

inputs of each branch are not identical. For a cropped patch, we transform it into the token through dimensionality reduction (e.g., FA) and a convolutional stem. In addition, the central pixel of the patch is extracted separately as a spectral vector. For the frequency feature learning branch, the input is the token. For the spectral feature learning branch, the input is the central pixel vector. For the spatial feature learning branch, the input is the token and the central pixel vector. Considering the different inputs and different learning perspectives of each feature extraction branch, we are convinced that the resulting multidomain features are informative enough to facilitate the classification performance.

Moreover, we construct the network architecture in a hierarchical manner to learn representations at different levels. Concretely, we set up convolutional layers at intervals within each branch, except for the spectral feature learning branch. Since the input size of a patch-based HSI classification framework is generally not very large, we determine to reduce the size by 2 at a time.

B. Frequency Feature Learning

Let the original HSI data be $\mathcal{H} \in \mathbb{R}^{h\times w\times b}$, where h, w, and b denote the height, width, and the number of spectral bands, respectively. Based on $\mathcal{H}$, the HSI classification task aims to assign a land cover category label to each pixel by exploiting the spectral and spatial information in $\mathcal{H}$. Moreover, HSI is characterized by abundant spectral bands, which, however, inevitably introduces redundant information and computational burden to classification models. Consequently, some methods [41], [42] consider dimensionality reduction techniques such as PCA to compress HSI data during the preprocessing stage. Nonetheless, our prior work [36] revealed that FA might be a more appropriate choice for dimensionality reduction of HSI data compared to PCA. Specifically, there are inherent correlations between the spectral bands, especially the neighboring spectral bands. PCA is prone to overlook this point and thus lacks the ability to discriminate between some similar land covers. In contrast, FA represents HSI data as a blend of common factors and individual factors, thus better capturing commonalities and variations in the relevant spectral bands. Based on the above analysis, we employ FA to transform $\mathcal{H}$ into $\mathcal{H}' \in \mathbb{R}^{h\times w\times c}$, where c represents the number of bands after FA. Then, centered on each pixel, neighbor patches are extracted with the size of $p \times p \times c$. The label of each patch is determined by the label of the central pixel. A padding operation will be performed if the central pixel is located at the edge of HSI. Each generated patch is first converted to the token with a convolutional stem (i.e., a 3-D convolutional layer and 2-D convolutional layer), which is denoted as $X_t \in \mathbb{R}^{p\times p\times c}$.

For the frequency feature learning branch, a peculiar module called spectral–spatial frequency modulator (SSFM) is proposed to learn features from the perspective of the frequency domain. The detailed structure of SSFM is shown in Fig. 3. Following the general structure of the transformer (i.e., normalization layer, token mixer, normalization layer, and a feedforward network), we mainly concentrate on the design of the token mixer.

SSFM employs the discrete Fourier transform (DFT). Let F represent the Fourier transform and F^{-1} represent the inverse Fourier transform. Given a sequence of real numbers x_n, where $n = 1, 2, \ldots, N$. DFT is defined as $X_l = \sum_{n=0}^{N-1} x_n e^{-i\omega ln}$, where X_l is a sequence of complex numbers in the frequency domain. Similarly, the inverse DFT is defined as $x_n = \sum_{l=0}^{L-1} X_l e^{i\omega ln}$. In addition, the fast Fourier transform (FFT) algorithm can be utilized to accelerate the computation of DFT, which reduces the computational complexity from $O(N^2)$ to $O(N\log N)$. The Fourier transform exhibits the valuable property that time-domain convolution is equivalent to frequency-domain multiplication. Therefore, the FFT can be employed to accelerate the calculation of

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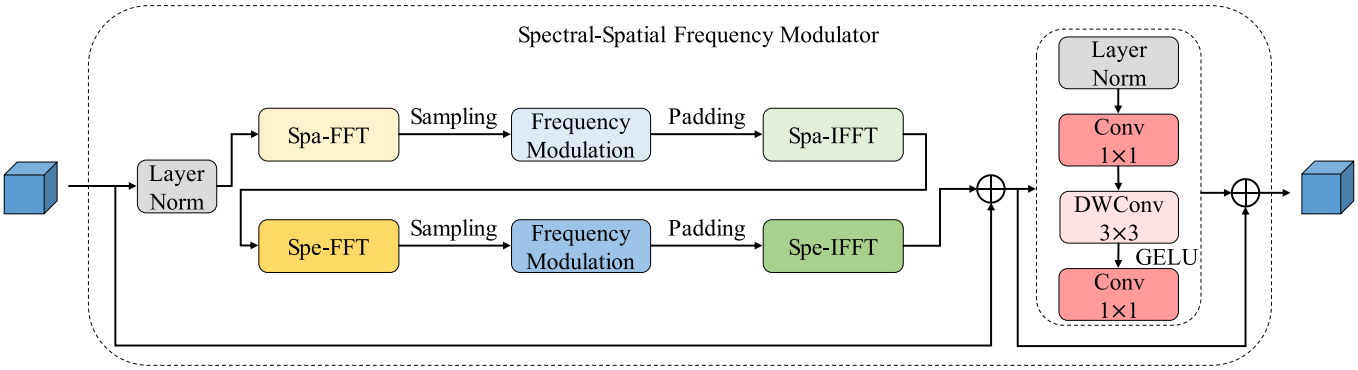


Fig. 3. Details of SSFM. Spa-FFT represents performing FFT on the spatial dimension and Spe-FFT stands for performing FFT on the spectral dimension.

CNN and serve as a neural operator. This Fourier-based neural operator is able to be wrapped as a block of the network to conduct efficient representation learning in the low-frequency domain [34]. In contrast, there is another research path, which is the random Fourier method using randomly selected modes [43], [44], [45].

Given the input token $X_t \in \mathbb{R}^{p\times p\times c}$, we adopt a three-step strategy (i.e., FFT, frequency modulation, and IFFT) to learn representations. For HSI classification, two subpatterns evolved from the abovementioned pattern are developed to enhance feature extraction along the spatial dimension and the spectral dimension. Concretely, we first reshape X_t into $\mathbb{R}^{n\times d}$, where $n = p \times p$. After that, we perform Spa-FFT (i.e., FFT acting on the first dimension of X_t) to convert X_t to the frequency domain. In the frequency domain, we randomly select m modes using a sampling strategy to retain. The process can be formulated as

$$Y_{\text{spa}} = \text{Select}\left(\mathcal{F}_{\text{spa}}(X_t)\right) \tag{2}$$

where $Y_{\text{spa}} \in \mathbb{C}^{m\times d}$ stands for the complex representation in the frequency domain and $m < n$. Note that although the Fourier basis is randomly selected, most of the information from $\text{F}(X_t)$ can be preserved [45].

Then, we conduct the frequency modulation to obtain the adjusted spectrum $\overline{Y_{\text{spa}}} \in \mathbb{C}^{m\times d}$

$$\overline{Y_{\text{spa}}} = FM\left(Y_{\text{spa}}\right) = Y_{\text{spa}} \odot R \tag{3}$$

where $R \in \mathbb{C}^{m\times d}$ is a parameterized kernel initialized with a standard normal distribution and $\odot$ is the Hadamard product.

Finally, the modulated spectrum $\overline{Y_{\text{spa}}}$ is zero-padded to $\mathbb{C}^{n\times d}$ and converted to the spatial domain with Spa-IFFT (i.e., the inverse FFT corresponding to Spa-FFT)

$$\overline{X_t} = \mathcal{F}_{\text{spa}}^{-1}\left(\text{Padding}\left(\overline{Y_{\text{spa}}}\right)\right) \tag{4}$$

where $\overline{X_t}$ denotes the updated token.

After we model the interactions between pixels within the token along the spatial dimension, we proceed to duplicate a similar operation along the spectral dimension. Similar to the above three-step strategy, the process along the spectral dimension can be formulated as

$$Y_{\text{spe}} = \text{Select}\left(\mathcal{F}_{\text{spe}}\left(\overline{X_t}\right)\right) \tag{5}$$

$$\overline{Y_{\text{spe}}} = FM\left(Y_{\text{spe}}\right) = Y_{\text{spe}} \odot R' \tag{6}$$

$$\widetilde{X_t} = \mathcal{F}_{\text{spe}}^{-1}\left(\text{Padding}\left(\widetilde{Y_{\text{spe}}}\right)\right) \tag{7}$$

where $\mathcal{F}_{\text{spe}}$ is Spe-FFT (namely, FFT acting on the second dimension of $\overline{X_t}$) and $\mathbb{R}' \in \mathbb{C}^{m\times d}$ is a kernel generated in the same way as R.

In addition, inspired by the success of CNNs, we integrate SSFM into the hierarchical architecture to construct the frequency feature learning branch, which is illustrated in the yellow box in Fig. 2.

C. Spectral Feature Learning

For pixel-level HSI classification, the spectral signature of the center pixel within the extracted patch plays an important role in determining the class label of the center pixel [38]. With this consideration in mind, we develop a spectral feature learning branch to value the influence of the spectral features associated with the center pixel.

In the spectral feature learning branch, the center pixel x_c of the original HSI patch is considered to provide spectral features. In HSIs, neighboring pixels in the spatial domain and neighboring bands in the spectral domain tend to be similar. Thus, discriminative information is overwhelmed by substantial spectral and spatial redundancy. Moreover, the high-dimensional characteristic of HSI data makes it challenging to concentrate on local nuanced features. In spectral feature learning, we develop the band grouping to safeguard significant correlations among neighboring bands, which is beneficial to retain essential discrimination information that contributes to the identification of certain classes. As shown in Fig. 4, we introduce the adaptive band grouping method to properly organize all the spectral bands. We achieve this goal based on the spectral correlation analysis. Fig. 5 displays the correlation coefficient curves of neighboring spectral bands, where there are some distinct valley points. These valley points suggest that the spectral bands are locally similar. Depending on these valley points, we can implement adaptive band grouping. Concretely, we use the Pearson correlation coefficients ρ to measure the spectral correlation

$$r(i) = \frac{\text{Cov}(s_i, s_{i+1})}{\sigma_i \sigma_{i+1}} \tag{8}$$

where s_i and s_{i+1} denote the ith and $(i+1)$th spectral bands, $\text{Cov}(\cdot)$ represents the covariance, and σ_i and σ_{i+1} mean the

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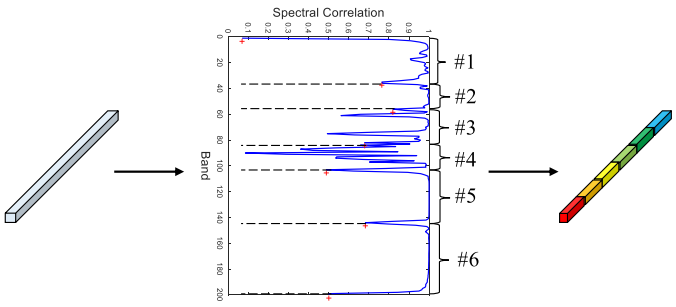


Fig. 4. Illustration of band grouping (taking the INs dataset as an example).

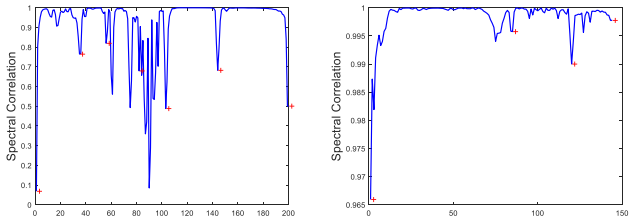


Fig. 5. Spectral correlation curves. (a) INs. (b) HU.

standard deviation of s_i and s_{i+1}, respectively. Then, the valley points are defined as follows:

$$\text{Index}(i) = \begin{cases} 1, & r(i) < \min\left(\bar{r}_l(i), \bar{r}_r(i)\right) \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where $\text{Index}(j) = 1$ stands for the valley point and $\bar{r}_l(i) = 1/N \sum_{n\in\Omega_l} r_n$ and $\bar{r}_r(i) = 1/N \sum_{n\in\Omega_r} r_n$ are the average values of correlation coefficients of N points located to the left and right of point i, respectively. In practice, N is fixed to 3 for all datasets. With the aid of these valley points, spectral bands are adaptively grouped.

For these band groups, we employ the spectral frequency modulator, the portion focused on the spectral dimension within the abovementioned SSFM, to model intragroup correlations.

D. Spatial Feature Learning

For patch-based HSI classification, there are two main sources of the most important information. One is the spectral domain with tens or even hundreds of bands, and the other is the spatial relationships between the center pixel and its neighboring pixels. In the popular patch-based pattern of HSI classification, a patch is used to represent its central pixel, and neighboring pixels to the center pixel serve as auxiliary roles. The center pixel, neighboring pixels, and pixels at a distance should not be treated the same. In addition, as shown in Fig. 1, when the center pixel of a patch is at the edge of a land cover class, the spatial neighboring information exerts a more significant influence on assigning labels since more interference information is included. With the aforementioned rationale in mind, we construct the spatial feature learning branch.

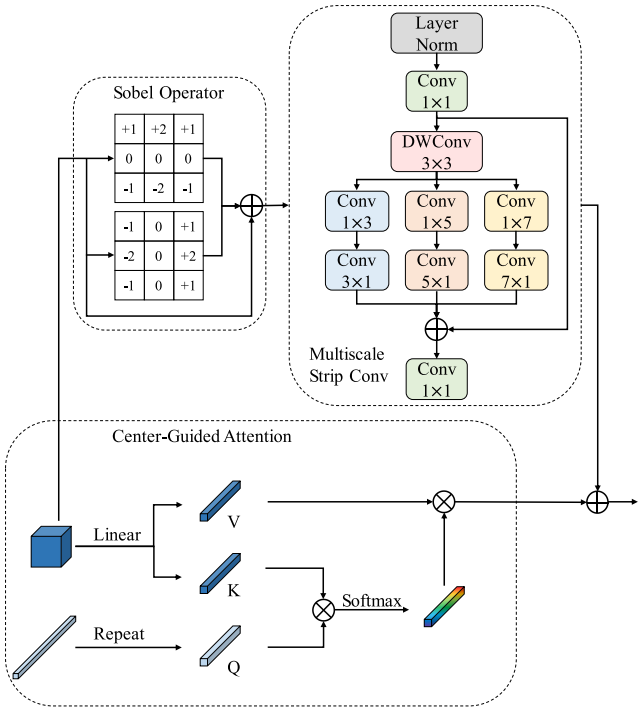


Fig. 6. Details of the proposed CEEM.

The important component of the spatial feature learning branch is the proposed center-edge enhanced module (CEEM), which is shown in Fig. 6. CEEM uses the token and the center pixel vector as input. First, we hope to highlight the central pixel and amplify the effect of neighboring pixels that highly resemble the central pixel. To this end, we design a mechanism named center-guided attention (CGA). Here, we introduce the center pixel vector to generate Q_i, while K_i and V_i are obtained by projecting the token

$$Q_i = q_i(\text{Repeat}(x_c)),\ K_i = k_i(X_t),\ V = v_i(X_t) \tag{10}$$

where $\text{Repeat}(\cdot)$ means expanding the 1-D center pixel vector x_c along the spatial dimension to the same spatial size as the input token X_t. $\{q_i(\cdot), k_i(\cdot), v_i(\cdot)\}$ are linear projections and $\{Q_i, K_i, V_i\}$ represent the query, key, and value for the ith head. The subsequent operations against $\{Q_i, K_i, V_i\}$ are similar to those in MHSA

$$Z_i = \text{Attention}(Q_i, K_i, V_i) = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d}}\right) V_i \tag{11}$$

$$\text{CGA}\left(X_t'\right) = \text{Concat}(Z_1, Z_2, \ldots, Z_i, \ldots, Z_s) W^O \tag{12}$$

where CGA also adopts the multihead mechanism such as MHSA.

Next, we introduce the Sobel operator to enhance edge information to mitigate the misclassification of pixels located at the edges. Specifically, we design two 3×3 kernels with fixed parameters and slide them like convolutions to implement the Sobel operator in horizontal and vertical directions. Two kernels are defined as

$$S_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix},\ S_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}. \tag{13}$$

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After that, we apply the two kernels to the input feature map F_i, generating the gradient maps M_x and M_y. The gradient maps are then fused with F_i to obtain edge-enhanced feature maps F_e

$$F_e = F_i \oplus M_{xy} \tag{14}$$

where $\oplus$ denotes the element-wise addition and M_{xy} is the concatenation of gradient maps along the spectral dimension.

To further explore the structural and shape information of HSI tokens, we design the multiscale strip convolution placed after the Sobel operator. As shown in Fig. 6, the proposed multiscale strip convolution consists of three main components: a depth-wise convolution, multibranch strip convolutions, and a 1×1 convolution. Mathematically, the process can be formulated as

$$F_{\text{out}} = \text{Conv}_{1\times 1}\left(F_{\text{in}} + \sum_{j=0}^{2} \text{Scale}_j\left(\text{Conv}_{\text{strip}}(F_{\text{in}})\right)\right) \tag{15}$$

where F_{in} and F_{out} denote the input and output feature map, respectively; $\text{Conv}_{\text{strip}}$ means the depth-wise strip convolutions and Scale_j; and $j \in \{0, 1, 2\}$ denotes the jth branch. In each branch, we use two depth-wise strip convolution with different kernels (e.g., 1×3 and 3×1). Moreover, the kernel size is also different for each branch, where it is set to 3, 5, and 7. The benefits of our design are manifold. First, the strip convolution is lightweight and two strip convolutions with different orientations can simulate a standard grid convolution [46]. Second, the strip convolution is less prone to introduce interference information than a standard grid convolution when the pixel is at the edge of a land cover region. Finally, according to the additivity of convolution, the superposition of two strip convolutions with different orientations is equivalent to a cross convolution, which facilitates the extraction of center pixel feature [42].

E. Multidomain Feature Fusion and Final Classification

As we use different branches and extractors to get multidomain features, the distributions of the features are different. In addition, we introduce the hierarchical structure for multiscale feature learning. As a result, the feature fusion is important. To better integrate the multidomain information of ground objects and avoid mutual interference, a weighted vector is proposed and dynamically assigned to features from different branches during the training process

$$F_{\text{weight}} = \text{Concat}\left(\lambda_1 \cdot F_{\text{fre}}, \lambda_2 \cdot F_{\text{spa}}, \lambda_3 \cdot F_{\text{spe}}\right) \tag{16}$$

where λ_i represents different weights and is dynamically assigned during model training; F_{weight} represents weighted outputs; and F_{fre}, F_{spa}, and F_{spe} are features from the frequency branch, spatial branch, and spectral branch, respectively.

To enjoy a better network training experience, we employ Polyloss [47] instead of cross-entropy (CE) loss. Polyloss can be considered as a linear combination of polynomial functions. In contrast to the widely used CE loss, Polyloss offers flexibility in adjusting the coefficients of various polynomial bases to align with the specific target task and dataset. Since CE loss is a special case of Polyloss, by fine-tuning the coefficients of polynomial bases, it is theoretically feasible to achieve improved results compared to using CE loss. In practice, adjusting the first polynomial term yields a significant gain, and hence, the simple Polyloss is adopted, which can be formulated as follows:

$$\mathcal{L}_{\text{Poly}} = \mathcal{L}_{\text{CE}} + \varepsilon(1 - p_t) \tag{17}$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss, ε is the polynomial coefficient, and p_t is the prediction probability of the target category label.

IV. Experiments and Analysis

A. Data Description

To test the performance of the proposed method, we use three public HSI datasets, including the Indian Pines (INs), Houston 2013 [48] (HU), and WHU-Hi-HanChuan [49](HC).

The IN dataset was acquired using the AVIRS HS sensor and showcases a farmland region situated in Northwestern Indiana. For the IN dataset, following a comprehensive correction process encompassing spectral, spatial, and atmospheric aspects, 200 spectral bands are retained for experiments. The selected area comprises 145×145 pixels and encompasses 16 distinct land cover classes.

The HU dataset, made available through the GRSS Data Fusion Contest 2013, originates from a scene captured by airborne sensors over the University of Houston campus and its surrounding areas in 2012. For the HU dataset, after the spectral, spatial, and atmospheric correction procedures, 144 spectral bands were preserved for experiments. The selected area encompasses 349×1905 pixels and includes 15 distinct land cover classes.

The HC dataset was acquired in 2016 by the Headwall Nano-Hyperspec sensor in Hanchuan, Hubei, China. This dataset features an image size of 610×340 pixels, encompassing 274 bands ranging from 0.4 to 1 μm, and includes 16 distinct land cover classes. Different from the previously mentioned datasets, which were collected using satellite or airborne hyperspectral platforms, the HC dataset stands out as it was captured employing a hyperspectral system mounted on an unmanned aerial vehicle (UAV). This unique approach provides high spatial resolution hyperspectral imagery.

A common way to divide the HSI datasets is the random sampling strategy. However, in the random sampling dataset, when neighborhood patches are extracted for the training set, these patches may overlap with the test information. To avoid the spatial overlapping phenomenon, we conduct our experiments on the datasets using the disjoint split strategy. It is worth mentioning that this dataset treatment is also commonly used in recent studies [26], [30], [50]. The numbers of training and test samples corresponding to the disjoint split strategy are reported in Table I.

B. Experimental Setup

To quantitatively assess the performance of the proposed model and compared methods, three commonly employed metrics in HSI classification are employed. These metrics

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THE INS, HU 2013, AND HCDATASET

Class No.	Color	Indian Pines			Houston 2013			WHU-Hi-HanChuan		
		Class name	Train	Test	Class name	Train	Test	Class name	Train	Test
1		Alfalfa	15	39	Healthy grass	198	1053	Strawberry	447	44288
2		Corn-notill	50	1384	Stressed grass	190	1064	Cowpea	227	22526
3		Corn-mintill	50	784	Synthetic grass	192	505	Soybean	102	10185
4		Corn	50	184	Trees	188	1056	Sorghum	53	5300
5		Grass-pasture	50	447	Soil	186	1056	Water spinach	12	1188
6		Grass-trees	50	697	Water	182	143	Watermelon	45	4488
7		Grass-pasture-mowed	15	11	Residential	196	1072	Greens	59	5844
8		Hay-windrowed	50	439	Commercial	191	1053	Trees	179	17799
9		Oats	15	5	Road	193	1059	Grass	94	9375
10		Soybean-notill	50	918	Highway	191	1036	Red roof	105	10411
11		Soybean-mintill	50	2418	Railway	181	1054	Gray roof	169	16742
12		Soybean-clean	50	564	Parking lot_1	192	1041	Plastic	36	3643
13		Wheat	50	162	Parking lot_2	184	285	Bare soil	91	9025
14		Woods	50	1244	Tennis court	181	247	Road	185	18375
15		Bldg-grass-tree-drives	50	330	Running track	187	473	Bright object	11	1125
16		Stone-steel-towers	50	45				Water	754	74647
Total			695	9671		2832	12197		2569	254961

include overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ). Note that a higher value for each of these metrics signifies a more favorable classification outcome.

Several recently advanced methods are selected for comparison. Concretely, four CNN-based methods are considered, which are SSRN [15], A2S2K [23], CDCNN [51], and DBDA [52]. Five transform-based methods, including SF [26], SSFTT [29], BS2T [30], ViT [53], and GMANet [54], are compared. The spatial size of each comparison method adheres to the specifications provided in their original papers. For CDCNN, the spatial size is set to 5×5. SSRN and SF use a spatial size of 7×7. DBDA, A2S2K, and BS2T employ a spatial size of 9×9. SSFTT uses a spatial size of 13×13. ViT is a pixel-wise method. For more in-depth information, we encourage referring to their respective original papers.

To mitigate the influence of random operations, each experiment is repeated ten times, and the average results along with the standard deviations of various metrics are reported. The proposed HSIC-SDM and other models used for comparison are implemented on the PyTorch platform. All the experiments are executed on a computer equipped with an Intel Core i7-8700K CPU, 32 GB RAM, an NVIDIA GeForce GTX 1080Ti GPU, and a 64-bit Windows 10 operating system. The adaptive moment estimation (Adam) optimizer [55] is used with a minibatch size of 64. The learning rate is initialized with 5e−4 and decayed by a factor of 0.95 after each epoch. The epoch is set to 200 and the early stopping technique is employed in the training process (i.e., the training process will be terminated if the loss function ceases to decrease within 20 epochs). To reduce memory usage and eliminate redundancy, we downscale all three datasets to 15-D by FA, as previously described.

C. Model Analysis

1) Ablation Analysis of Different Branches: The proposed HSIC-SDM comprises three critical branches: frequency feature learning, spatial feature learning, and spectral feature learning. In this section, we empirically assess the impact and effectiveness of these three branches.

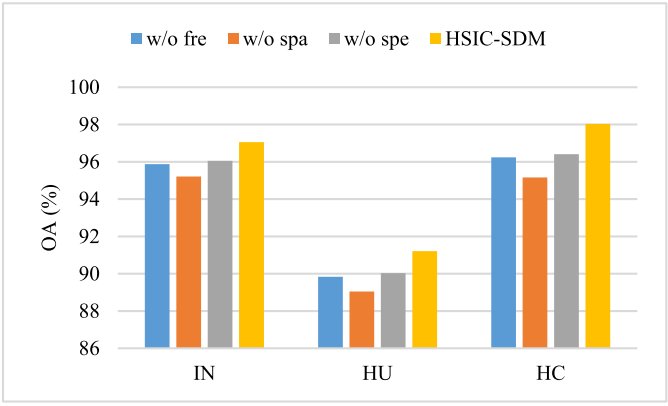


Fig. 7. Ablation study results on the different branches. "w/o" represents "without.".

To conduct our analysis, we systematically exclude one component at a time from the HSIC-SDM and denote the resulting versions as "w/o fre," "w/o spa," and "w/o spe." The key metric OA is adopted to evaluate the classification performance of different versions across three benchmark datasets. To visually compare specific OA values, we present a histogram in Fig. 7. Bars with distinct colors correspond to different versions of HSIC-SDM. The yellow bars represent the OA value of the complete HSIC-SDM, while the bars of other colors represent the OA value of variants with a specific branch removed. Remarkably, regardless of the dataset, the yellow bar consistently exhibits higher OA values than the other bars. This observation indicates that the performance of HSIC-SDM suffers to some degree when a branch is removed from it. Moreover, the orange bar always shows the lowest OA value, which implies that the spatial feature learning branch may be the most influential. In a nutshell, we can conclude that all three branches contribute significantly to the excellent performance of HSIC-SDM.

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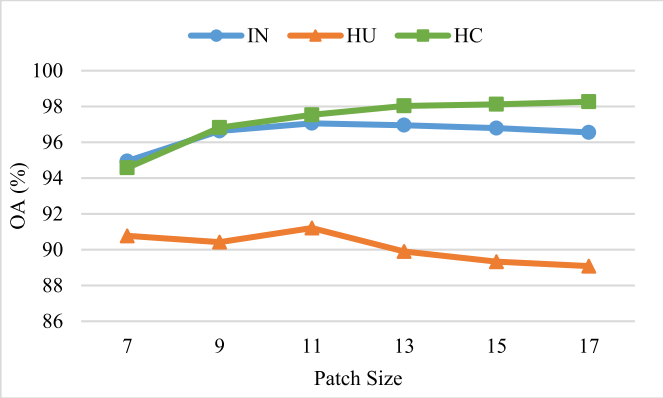


Fig. 8. OA (%) of HSIC-SDM with different patch sizes on three datasets.

2) Ablation Analysis of Patch Size: It is well known that smaller patch sizes lead to less valuable information, while larger patch sizes introduce more potential interference information. Therefore, it becomes imperative to determine the optimal patch size for the patch-based HSI classification framework. To this end, we investigate the impact of varying patch sizes on the classification and plot the results in Fig. 8.

As illustrated in Fig. 8, for different HSIs, the OA values of HSIC-SDM exhibit different trends with changing patch sizes. Specifically, for the IN and HU datasets, the OA values roughly increase and then decrease as the patch size grows, and the OA values of HSIC-SDM on the two datasets peak at a patch size of 11. For the HC dataset, the OA value demonstrates a steady increase with an expanding patch size, but such a gain becomes less significant when the patch size exceeds 13. The above phenomenon can be explained as follows. In terms of the IN and HU datasets, land cover distribution consists of small areas, scattered points, or lines. Therefore, the patch sizes suitable for the two datasets should not be too large; otherwise, more complex structures and more pixels from different classes would be contained. On the other hand, the HC dataset, characterized by large and continuous land cover areas, benefits from larger patch sizes compared to the previous two. Considering the balance between accuracy and computational cost, we set the patch size as 11 for the IN and HU datasets and 13 for the HC dataset.

3) Ablation Analysis of Dimension Reduction: Effective feature preprocessing is crucial for reducing redundant information and lightening the computational burden. As previously noted, we opt for FA as an alternative to the commonly employed PCA for dimensionality reduction. We evaluate the performance of the proposed HSIC-SDM under three scenarios, i.e., without dimensionality reduction, using PCA, and using FA. The results are reported in Table II. As indicated in Table II, the model employing FA delivers the best performance and is quite time-efficient. In contrast, the model without dimensionality reduction significantly lags behind the other two alternatives in terms of time complexity and OA scores. In addition, there remains a notable gap in OA between the model with FA and the model with PCA. This disparity can be attributed to FAs suitability for HSI data, particularly in capturing variations between relevant spectral

TABLE II
IMPACT OF THE PROPOSED METHOD WITH DIFFERENT DIMENSION REDUCTIONS ON THREE DATASETS

Dimension Reduction	Metric	IN	HU	HC
No	OA (%)	95.86	86.03	94.11
	Test.(s)	3.31	2.89	170.39
PCA	OA (%)	94.47	88.78	95.68
	Test.(s)	1.13	1.40	43.84
FA	OA (%)	97.06	91.21	98.03
	Test.(s)	1.23	1.47	45.50

TABLE III
IMPACT OF DIFFERENT NUMBERS OF BANDS AFTER FA ON THREE DATASETS

Band Number	Metric	IN	HU	HC
10	OA (%)	95.96	90.62	97.08
	Test.(s)	1.19	1.35	44.97
15	OA (%)	97.06	91.21	98.03
	Test.(s)	1.23	1.47	45.50
30	OA (%)	96.98	89.76	98.18
	Test.(s)	1.25	1.55	73.71
60	OA (%)	96.91	88.92	98.37
	Test.(s)	1.55	1.67	89.22

bands. It is worth noting that on the IN dataset, the OA score without dimensionality reduction is even higher than the OA score based on PCA dimensionality reduction. Upon close examination of the experimental results, we observe a significant decline in the accuracy of categories 2, 10, and 11 when PCA is applied. This may suggest that PCA led to the loss of significant spectral information crucial for identifying these three categories.

Furthermore, we performed experiments to ascertain the optimal number of bands for patches after using FA. The results are detailed in Table III. As shown in Table III, on all three datasets, the test time rises with the number of bands, while the OA score initially increases and then decreases. Striking a balance between accuracy and running speed, we ultimately elected to retain 15 bands after applying FA.

4) Impact of Loss Function and Sample Ratio: For the loss function, we adopted the Polyloss loss function instead of the regular CE loss function. In addition, in CEEM, we preserved components in the frequency domain with sampling to reduce redundancy. Here, we empirically investigate the effects of both and the results are shown in Fig. 9.

As demonstrated in Fig. 9, the switch to the Polyloss holds promise. Compared to the CE loss, Polyloss leads to an improvement in OA scores for different sampling ratios on all three datasets. It can be observed from Fig. 9 that for each dataset, varying sampling ratios introduce subtle fluctuations in OA scores. Moreover, there exists a sampling ratio that maximizes the OA score. Specifically, this optimum sampling ratio is determined as 0.5 for IN, 0.7 for HU, and 0.5 for HC.

5) Ablation Analysis of CEEM: In CEEM, we emphasize the significance of central pixel and edge information by employing center enhancement and edge enhancement, respectively. To identify an appropriate construction paradigm for CEEM, we design four similarity measures related to the

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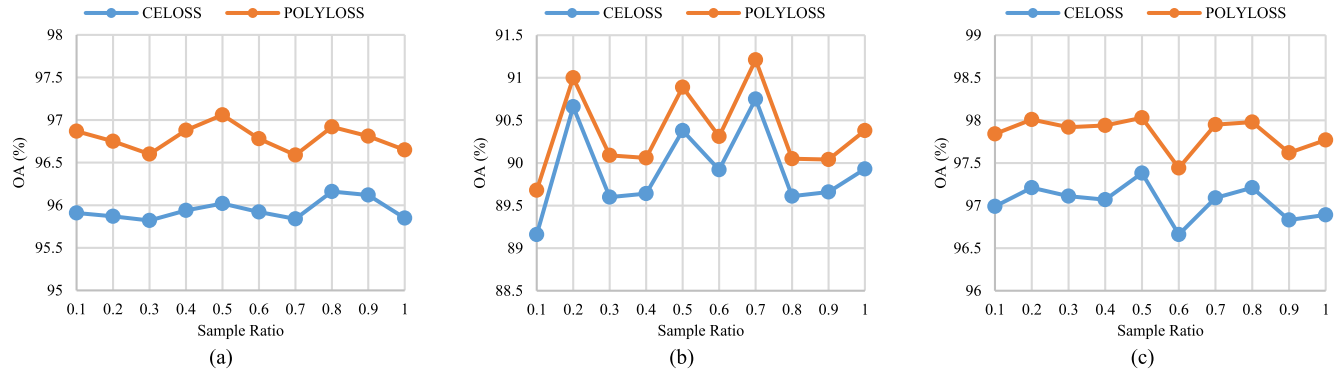


Fig. 9. Impact of different loss functions and sample ratios. (a) IN dataset. (b) HU dataset. (c) HC dataset.

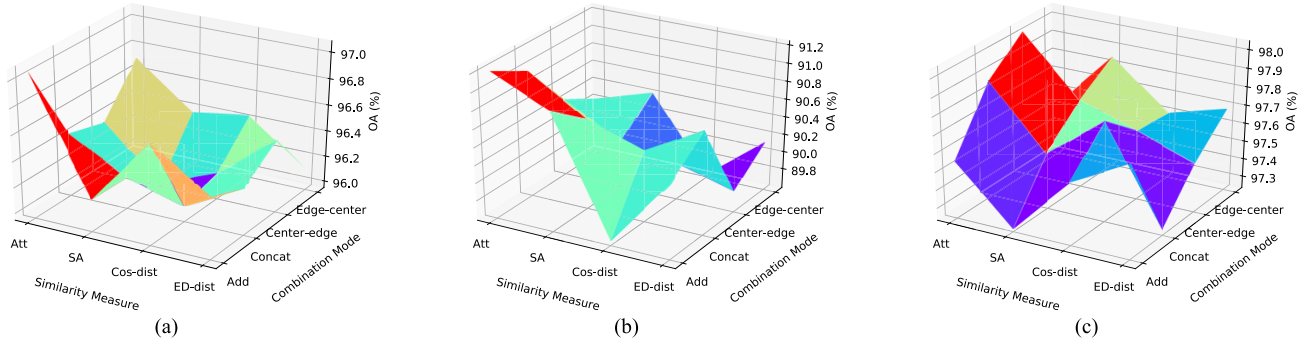


Fig. 10. Impact of different similarity measures and fusions. (a) IN dataset. (b) HU dataset. (c) HC dataset.

center pixel and four combination modes for two enhancements. Concretely, in terms of similarity measure, we offer four distinct options, namely, "Att," "SA," "Cos-dist," and "ED-dist." "Att" implies attention, i.e., using the original central pixel vector to perform the attention mechanism with the feature map. "SA" means self-attention, i.e., using the feature map to perform the attention mechanism with itself. "Cos-dist" and "ED-dist" stand for cosine distance and Euclidean distance, respectively, which are used to calculate the similarity between the original central pixel vector and the feature map. Moreover, in terms of the combination mode for two enhancements, we provide four choices, namely, "Add," "Concat," "Center-edge," and "Edge-center." "Add" means to add the outputs of both enhancements. "Concat" refers to concatenating the outputs of both enhancements. "Center-edge" and "Edge-center" indicate the specific order in which the two enhancements are connected in series.

As depicted in Fig. 10, the OA scores fluctuate with different similarity measures and combination modes. Different from the other three similarity measures, "SA" only utilizes the intermediate feature map and does not perform as well as the other three for the most part. For all three datasets, it appears that the similarity measure of "Att" is superior, while the best combination mode varies. To achieve the best performance, "Att" is chosen as the enhancement for the central pixel on all three datasets. As for the combination mode, "Add" is adopted for the IN and HU datasets, while "Center-edge" is the preferred choice for the HC dataset.

D. Classification Results

In this section, we present a comprehensive comparison between our proposed method and several classical and SOTA methods, evaluating them both quantitatively and qualitatively. Please note that our dataset configurations for the IN and HU datasets are different from those used in the original papers of most comparison methods. Consequently, the results we reproduce may exhibit disparities from those reported in the original papers.

1) Quantitative Analysis: Tables IV–VI report the average values and standard deviations of OA, AA, Kappa coefficient, and class-specific accuracies acquired by different methods on the IN, HU, and HC datasets, respectively. The highest values are denoted in bold, while the second-highest values are underlined for ease of reference.

Table IV shows that the latest CNN-based models or transformer-based models demonstrate favorable performance. Nevertheless, the proposed model surpasses all the compared methods, primarily attributed to its greater ability to learn spatial and spectral information from multiple perspectives. HSIC-SDM achieves impressive results with OA, AA, and Kappa values of 97.06%, 98.43%, and 96.63%, respectively. These figures are notably higher than the second-ranked A2S2K, with margins of 1.07%, 0.57%, and 1.22%, respectively. In addition, HSIC-SDM delivers commendable performance in terms of class-specific accuracy. It secures first place in 10 of the 16 categories and takes second place in 2 of the 16 categories.

Table V presents the classification results on the HU dataset. Most of the compared methods, except for a few classical

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TABLE IV
CLASSIFICATION PERFORMANCE BY DIFFERENT METHODS FOR IN DATASET

Class	CDCNN	SSRN	DBDA	A2S2K	ViT	SF	SSFTT	BS2T	GMANet	HSIC-SDM
1	26.66±14.46	98.82±0.38	97.69±0.76	99.74±0.56	81.53±5.47	77.69±6.59	98.97±1.25	97.43±1.65	**100±0**	**100±0**
2	54.00±3.80	92.02±1.04	91.37±1.91	**94.23±1.09**	67.21±7.20	72.95±2.22	82.19±1.92	90.23±2.85	93.96±4.82	93.27±0.38
3	48.72±11.55	98.06±0.47	98.58±0.54	97.94±0.96	79.31±3.91	86.28±2.98	98.11±0.27	96.07±1.58	99.31±0.35	**99.90±0.15**
4	96.08±0.79	99.21±0.08	**100±0**	**100±0**	95.76±1.98	97.22±1.67	98.80±0.63	**100±0**	99.89±0.04	**100±0**
5	84.69±1.91	95.64±0.64	92.32±0.34	95.52±0.50	93.42±1.05	95.32±1.39	95.92±1.12	92.08±1.47	95.88±0.97	**97.45±0.89**
6	91.96±1.83	98.49±0.21	99.55±0.26	**99.79±0.29**	91.44±4.41	93.80±2.06	95.17±0.76	99.34±0.02	96.32±1.21	98.94±0.82
7	89.09±8.90	**100±0**	**100±0**	**100±0**	90.90±9.95	99.09±2.72	**100±0**	**100±0**	**100±0**	**100±0**
8	98.26±0.88	98.88±0.11	**100±0**	99.86±0.11	96.62±1.69	97.69±1.51	99.86±0.18	99.86±0.01	99.41±0.18	**100±0**
9	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**
10	73.07±3.52	96.01±1.81	96.24±1.72	93.47±1.84	80.34±3.69	82.49±2.41	89.84±1.54	93.63±2.53	95.01±1.76	**97.08±1.57**
11	62.61±3.07	94.39±1.62	92.22±1.25	93.42±1.98	62.48±4.26	67.11±4.22	78.32±1.27	93.24±1.84	92.47±2.07	**95.91±1.08**
12	42.02±4.13	93.50±1.09	93.42±1.02	92.51±1.17	67.05±2.89	70.24±3.19	90.35±1.52	**95.00±0.66**	91.91±1.54	94.01±0.73
13	99.50±0.24	99.88±0.05	**100±0**	**100±0**	99.13±0.30	99.75±0.30	**100±0**	**100±0**	99.87±0.11	99.38±0.51
14	80.08±1.34	98.85±0.16	98.03±0.27	**99.38±0.25**	91.43±2.55	92.18±2.12	98.72±0.47	98.47±0.39	98.95±0.29	99.13±0.64
15	80.30±3.11	99.24±0.11	**100±0**	99.96±0.09	87.87±3.87	94.72±1.63	99.39±0.93	**100±0**	99.15±0.21	99.94±0.10
16	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**
OA(%)	68.98±1.31	95.88±0.52	95.31±0.61	95.99±0.49	77.69±1.74	81.29±1.44	89.59±0.48	95.07±0.52	95.60±1.15	**97.06±0.28**
AA(%)	76.69±1.78	97.98±0.20	97.46±0.29	97.86±0.17	86.53±1.33	89.16±0.86	95.35±0.42	97.21±0.21	97.63±0.78	**98.43±0.05**
Kappa	64.74±1.49	95.28±0.59	94.64±0.70	95.41±0.56	74.76±1.98	78.83±1.60	88.16±0.55	94.35±0.87	94.96±1.30	**96.63±0.41**

TABLE V
CLASSIFICATION PERFORMANCE BY DIFFERENT METHODS FOR HU DATASET

Class	CDCNN	SSRN	DBDA	A2S2K	ViT	SF	SSFTT	BS2T	GMANet	HSIC-SDM
1	81.93±0.26	83.06±0.06	82.33±0.31	82.98±0.12	82.08±0.29	81.55±0.35	83.01±0.07	82.52±0.13	81.80±0.89	**83.10±0.26**
2	82.71±0.94	85.03±0.20	85.12±0.04	86.84±4.43	83.61±0.49	84.48±0.41	94.07±7.21	85.02±0.52	85.15±3.21	**95.58±0.44**
3	90.77±0.83	99.82±0.21	**99.86±0.15**	99.34±0.48	99.60±0.16	71.92±2.31	99.60±0.25	99.45±0.04	95.96±1.07	99.80±0.12
4	92.78±0.34	94.50±2.18	92.38±0.21	**94.58±2.88**	91.95±0.63	90.83±0.59	90.43±2.95	91.95±1.83	91.89±2.33	93.08±0.73
5	99.77±0.14	**100±0**	**100±0**	**100±0**	97.56±0.29	98.28±0.57	**100±0**	**100±0**	**100±0**	**100±0**
6	94.96±2.77	97.41±1.74	97.97±1.93	96.43±1.34	94.87±0.33	90.62±3.55	94.40±1.71	97.34±0.62	99.02±0.18	**100±0**
7	80.20±1.01	84.45±2.80	83.25±2.10	**87.14±2.10**	72.51±2.16	78.81±2.64	80.24±1.10	85.15±2.37	79.49±0.95	80.13±1.01
8	61.61±5.41	62.47±12.20	66.00±9.38	69.57±10.33	42.76±1.51	60.55±7.30	78.74±2.78	73.12±0.44	57.32±6.59	**80.51±0.72**
9	77.67±2.91	85.84±2.77	**92.05±0.88**	82.97±4.76	70.82±0.74	73.47±2.99	83.98±2.81	87.19±0.95	89.97±3.72	87.53±0.54
10	55.27±5.51	53.16±3.31	57.19±4.89	53.37±6.55	46.23±4.62	46.24±2.44	60.86±5.81	59.84±5.12	53.18±5.58	**86.44±1.67**
11	68.48±2.38	83.82±2.15	92.25±6.46	83.18±2.84	67.93±1.01	59.75±4.54	**97.02±1.52**	90.77±0.76	92.92±4.10	96.30±1.21
12	84.74±1.74	96.16±2.46	97.95±0.52	92.69±4.71	52.61±4.87	82.22±1.99	**98.21±0.51**	98.00±0.17	91.72±2.64	98.08±0.16
13	89.82±0.80	85.36±3.81	85.15±4.03	**92.66±0.33**	58.24±4.16	64.56±3.53	89.96±0.61	70.94±1.23	75.78±4.21	92.28±0.43
14	94.73±0.44	**100±0**	**100±0**	**100±0**	**100±0**	84.49±1.73	**100±0**	**100±0**	**100±0**	**100±0**
15	92.01±1.44	99.91±0.06	99.49±0.90	99.28±1.25	95.42±0.81	79.38±6.56	99.87±0.25	**100±0**	98.98±0.11	**100±0**
OA(%)	80.37±1.01	84.83±0.90	86.54±1.15	85.37±1.28	73.61±1.08	75.76±0.94	88.17±1.54	86.63±0.88	84.99±0.92	**91.21±0.42**
AA(%)	83.16±0.77	87.40±0.67	88.73±1.03	88.07±1.06	77.08±1.02	76.48±1.03	89.93±1.23	88.08±0.75	86.71±0.78	**92.79±0.30**
Kappa	78.81±1.08	83.62±0.96	85.44±1.23	84.19±1.36	71.61±1.18	73.81±1.01	87.15±1.67	85.56±1.03	83.65±1.15	**90.45±0.51**

ones such as the CNN-based CDCNN and transformer-based ViT and SF, exhibit similar performance. In contrast, the proposed method clearly stands out with significantly better results. Specifically, HSIC-SDM achieves OA, AA, and Kappa values of 91.21% (±0.42%), 92.79% (±0.30%), and 90.45% (±0.51%), respectively, outperforming the second-ranked SSFTT by margins of 3.04%, 2.86%, and 3.3%, respectively.

Table VI provides the classification results on the HC dataset. Notably, SSRN, DBDA, A2S2K, SSFTT, BS2T, and GMANet show high OA values, yet their AA values are considerably lower compared to their OA values. The reason for this discrepancy may be that these models may perform poorly in some categories. A closer look at Table VI reveals that the categories responsible for these differences are primarily 5, 6, 13, and 15. In contrast, HSIC-SDM consistently demonstrates significantly better accuracy in these four categories than the compared methods. Consequently, the OA and AA of HSIC-SDM surpass those of its competitors by a considerable margin.

2) Qualitative Analysis: We conduct a qualitative evaluation by visualizing the classification maps generated by different methods. Figs. 11–13 illustrate the classification maps for three datasets. For the IN dataset, the classification maps of SSRN, DBDA, A2S2K, BS2T, GMANet, and HSIC-SDM exhibit better alignment with the ground truth and the delineation of edges is clearer than the other methods. Among these six methods, the edges of the classification map obtained by HSIC-SDM stand out with more prominent and ground-truth-consistent edges, reflecting its improved ability to learn edge information. For the HU dataset, the classification map produced by HSIC-SDM displays fewer instances of noise and classification errors, which implies that the proposed method captures finer details and alleviates excessive edge smoothing. Similarly, for the HC dataset, the classification maps generated by the compared methods tend to exhibit misclassified points along with noticeable noises, particularly in small areas (e.g., the pink region in the middle of the classification map). In contrast, the proposed method yields fewer misclassified points, even in small regions. This can be attributed to the fact

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TABLE VI
CLASSIFICATION PERFORMANCE BY DIFFERENT METHODS FOR HC DATASET

Class	CDCNN	SSRN	DBDA	A2S2K	ViT	SF	SSFTT	BS2T	GMANet	HSIC-SDM
1	95.89±0.45	98.43±0.41	**99.19±0.30**	98.88±0.19	97.22±0.38	96.47±0.23	96.95±0.59	98.08±0.42	99.19±0.21	99.11±0.71
2	86.45±3.73	90.95±2.92	96.23±1.43	96.25±0.41	90.83±1.68	91.27±1.31	90.88±1.41	94.97±0.98	98.12±0.38	**99.04±0.63**
3	52.52±29.60	93.15±1.74	96.18±1.75	95.64±0.86	86.32±6.62	94.26±0.36	87.35±1.29	97.32±1.25	**99.96±0.10**	99.09±0.24
4	72.52±16.77	97.19±0.64	96.49±0.27	96.57±0.49	95.82±1.63	95.53±0.27	97.21±0.54	94.86±0.80	94.24±0.49	**99.72±0.41**
5	0±0	43.32±32.35	78.79±6.12	55.58±39.31	57.93±26.41	69.97±5.17	63.62±3.29	81.90±4.28	0±0	**90.13±1.58**
6	1.88±2.66	75.68±2.32	68.87±5.11	71.22±7.45	30.89±9.23	47.05±4.98	63.29±2.40	60.43±4.57	65.22±8.75	**87.88±1.82**
7	82.11±5.52	91.18±2.61	89.84±2.52	92.06±0.13	84.31±3.92	83.87±6.95	89.68±1.11	85.45±2.89	**93.01±3.92**	92.11±0.45
8	64.32±5.58	91.78±3.55	91.88±2.78	92.17±1.09	77.41±7.14	78.58±1.57	88.12±2.08	83.65±1.41	91.56±0.87	**97.49±0.62**
9	46.51±0.54	92.01±2.96	94.77±2.48	92.18±0.96	75.80±6.97	82.57±0.64	85.63±3.11	88.94±2.63	95.56±0.53	**96.81±0.59**
10	83.75±2.31	96.98±0.65	97.76±1.07	97.38±0.45	92.49±4.99	96.63±0.73	96.99±1.04	97.34±0.94	94.48±3.14	**98.82±0.23**
11	82.11±5.75	98.52±0.63	90.76±8.39	**98.67±0.38**	90.13±7.47	92.52±2.23	96.95±0.53	94.22±0.85	96.02±0.64	98.62±0.17
12	1.45±2.05	75.13±5.96	61.92±21.2	86.39±5.26	53.19±21.19	74.16±7.19	65.14±7.66	79.47±4.14	65.96±5.76	**91.47±1.66**
13	39.14±1.90	78.78±2.83	80.23±1.76	80.34±1.44	71.25±3.46	75.96±1.96	75.16±4.03	84.47±1.19	83.67±4.91	**89.93±0.72**
14	88.85±2.03	92.79±0.84	92.85±0.95	95.80±0.48	91.77±0.97	93.85±0.32	91.95±0.63	94.48±1.10	95.21±0.75	**97.86±0.48**
15	58.42±10.79	71.76±3.95	72.21±4.36	**85.36±3.39**	23.72±14.01	43.56±18.77	65.24±6.78	81.45±5.72	0±0	83.41±2.51
16	98.55±0.48	98.56±1.54	98.54±0.32	99.80±0.04	99.39±0.33	99.47±0.22	99.57±0.13	99.66±0.11	99.55±0.09	**99.81±0.08**
OA(%)	81.75±1.64	94.44±1.17	94.65±1.53	95.87±0.53	90.08±2.25	92.02±0.67	93.13±0.34	94.44±0.59	95.34±2.10	**98.03±0.28**
AA(%)	59.65±3.79	86.64±2.73	87.91±2.42	89.64±3.55	76.15±6.11	82.23±2.22	84.61±1.09	88.54±1.71	79.78±2.96	**95.08±0.75**
Kappa	78.53±1.98	93.49±1.37	93.74±1.79	95.29±0.63	88.36±2.65	90.65±0.78	91.95±0.39	93.49±0.73	94.54±1.87	**97.76±0.34**

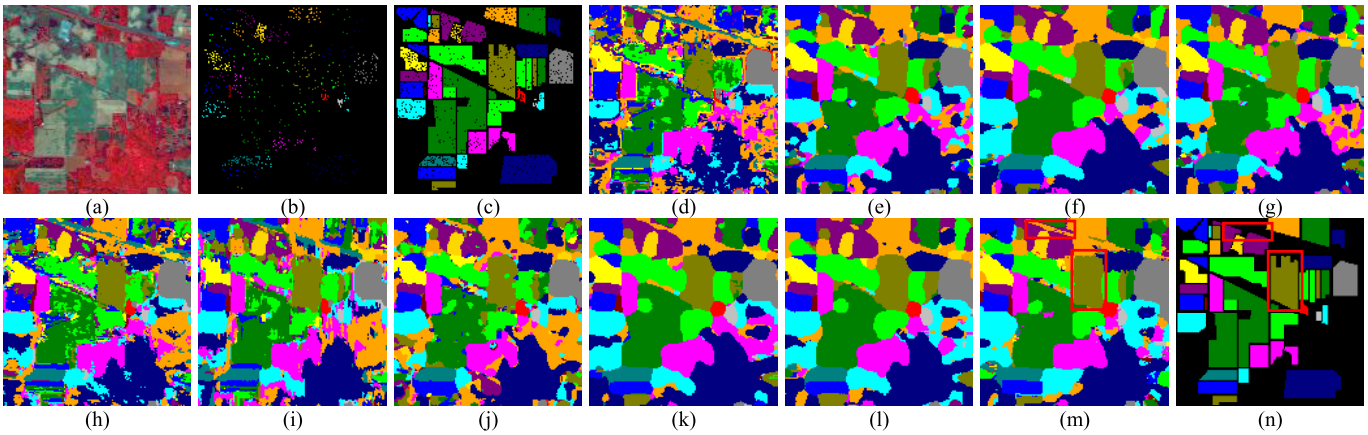


Fig. 11. Classification maps for the IN dataset. (a) False-color image. (b) Training. (c) Testing. (d) CDCNN. (e) SSRN. (f) DBDA. (g) A2S2K. (h) ViT. (i) SF. (j) SSFTT. (k) BS2T. (l) GMANet. (m) HSIC-SDM. (n) Ground truth.

that HSIC-SDM can effectively model short- and long-range dependencies and utilize multidomain features, resulting in significantly improved classification results for categories with finite samples. Moreover, we highlight representative regions in red or yellow boxes to illustrate the proposed model's ability to preserve edges and its capacity to recognize slim targets.

Fig. 14 presents the results of T-stochastic neighbor embedding (SNE) visualization of hidden features obtained by different DL methods on three datasets. Compared with competing methods, a noticeable distinction can be observed that our proposed HSIC-SDM exhibits a more robust aggregation distribution, characterized by reduced cross-aggregation and misclassification among categories. In particular, the scatters of similar features generated by the proposed HSIC-SDM are more concentrated, while the central clusters of different feature types are more distinctly separated. This observation indicates that the proposed HSIC-SDM effectively models various dependencies of different land covers in HSIs and exhibits a superior capacity to learn representations from land cover categories that are challenging to distinguish.

E. Discussion

1) Analysis of Model Complexity: To investigate the complexity and efficiency of the proposed HSIC-SDM, we provide a comparison of various performance metrics, including training time, testing time, trainable parameters, and floating-point operations (FLOPs), for SSRN, DBDA, BS2T, and HSIC-SDM. The results are summarized in Table VII. Table VII indicates that HSIC-SDM generally necessitates fewer parameters and FLOPs compared to the other methods. In addition, HSIC-SDM does not show a significant advantage in terms of training time over its competitors. This may be due to the fact that it employs multidomain features, thus requiring more time to acquire stable representations. Nevertheless, HSIC-SDM demonstrates superior efficiency with respect to testing time, which suggests that HSIC-SDM is suitable for practical applications.

2) Analysis of Stability and Robustness: To assess the stability and robustness of the proposed model, we randomly select training samples from the used training samples at different percentages on the IN dataset. Specifically, we choose the training percentage in 10% increments, ranging from 10%

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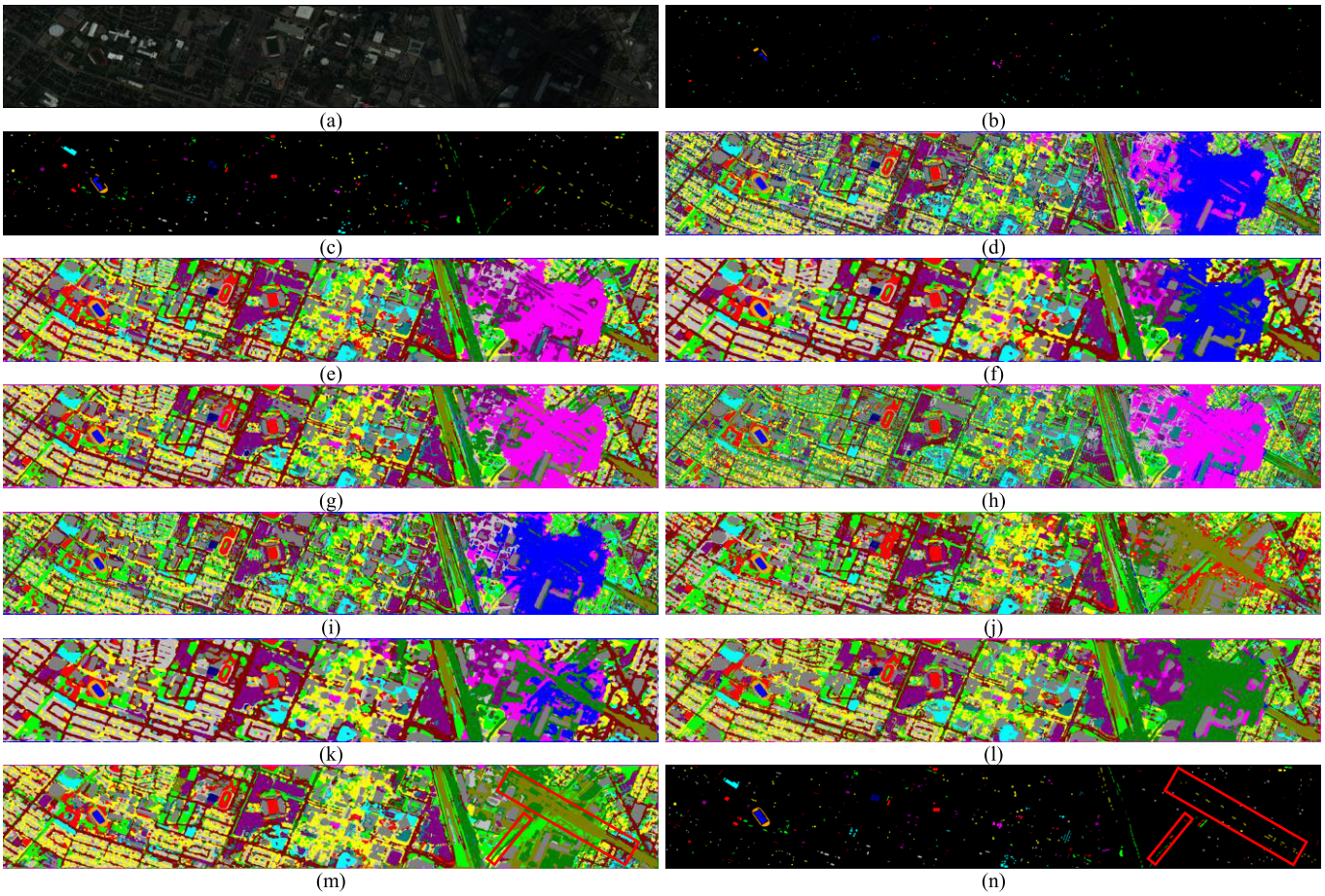


Fig. 12. Classification maps achieved by different methods for the HU dataset. (a) False-color image. (b) Training. (c) Testing. (d) CDCNN. (e) SSRN. (f) DBDA. (g) A2S2K. (h) ViT. (i) SF. (j) SSFTT. (k) BS2T. (l) GMANet. (m) HSIC-SDM. (n) Ground truth.

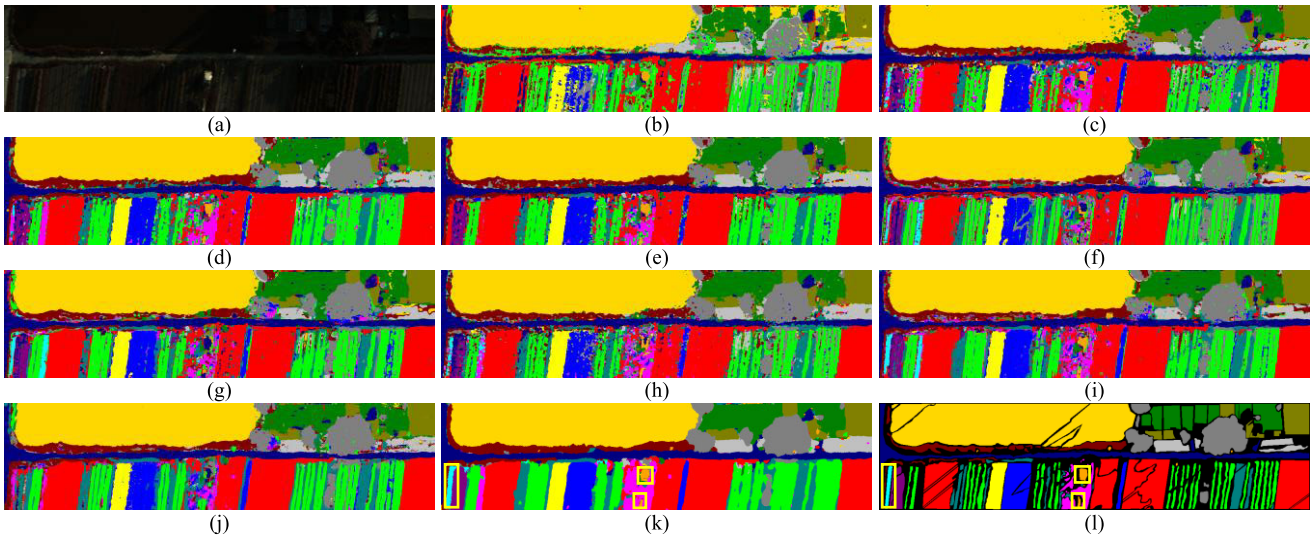


Fig. 13. Classification maps achieved by different methods for the HC dataset. (a) False-color image. (b) CDCNN. (c) SSRN. (d) DBDA. (e) A2S2K. (f) ViT. (g) SF. (h) SSFTT. (i) BS2T. (j) GMANet. (k) HSIC-SDM. (l) Ground truth.

to 100%. For each training percentage, we repeat the experiment five times and record the average OA. Fig. 15 presents the average OA scores obtained by DBDA, BS2T, and HSIC-SDM. As shown in Fig. 15, the OA scores exhibit a consistent and reasonable trend, i.e., the classification performance gradually improves as the percentage of used training samples increases. Notably, HSIC-SDM consistently outperforms the other two methods. This advantage is particularly prominent

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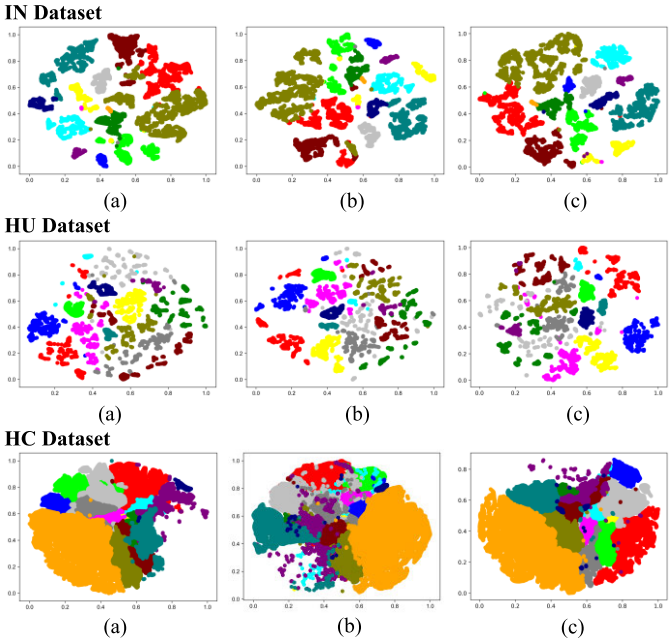


Fig. 14. t-SNE visualization of the hidden features in the latent space generated by (a) DBDA, (b) BS2T, and (c) HSIC-SDM on three datasets. The corresponding OA scores are listed in the captions.

TABLE VII
RUNNING TIME AND COMPLEXITY OF DIFFERENT METHODS ON THREE DATASETS

Method	Metric	IN	HU	HC
SSRN	Train. (s)	72.77	132.07	178.91
	Test. (s)	3.09	2.83	108.94
	Params (M)	0.364	0.278	0.478
	FLOPs (M)	157.423	112.680	216.548
DBDA	Train. (s)	86.95	215.35	232.55
	Test. (s)	4.62	4.33	164.26
	Params (M)	0.382	0.280	0.517
	FLOPs (M)	105.753	75.625	145.565
BS2T	Train. (s)	104.16	283.66	305.94
	Test. (s)	5.36	5.25	189.43
	Params (M)	0.277	0.274	0.512
	FLOPs (M)	105.844	75.716	145.656
HSIC-SDM	Train. (s)	84.50	335.21	257.77
	Test. (s)	1.23	1.47	45.50
	Params (M)	0.323	0.311	0.339
	FLOPs (M)	33.613	32.712	51.539

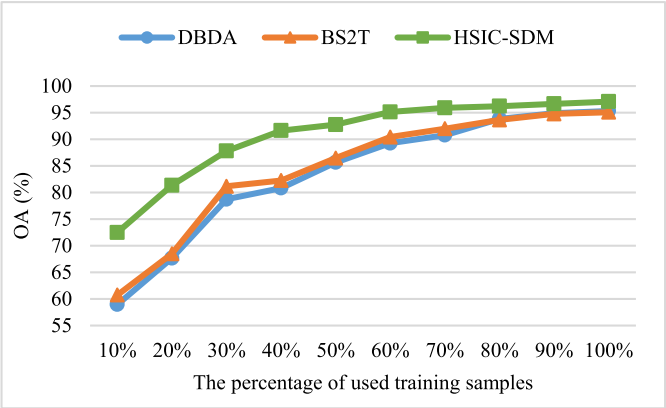


Fig. 15. OA with the varying number of used training samples generated by DBDA, BS2T, and HSIC-SDM on the IN dataset.

when the training data are scarce (e.g., 10% and 20%). Overall, our method is characterized by excellent stability and robustness.

V. CONCLUSION

In this article, we conduct an in-depth analysis of most existing methods that do not adequately leverage and exploit the characteristics of HSIs and the patch-based HSI classification pattern. We conclude the deficiencies into three problems. To alleviate these problems, we propose an HSIC-SDM. The proposed model utilizes two types of inputs and three branches to comprehensively exploit the information of HSIs under the patch-based HSI classification pattern. In particular, apart from the HSI patches extracted around the central pixel, the central pixel vector of the original HSI is also included as the input to the proposed model, aiming to provide complete spectral information for learning. Based on these two types of inputs, three branches are designed to learn multidomain features from diverse perspectives. As a response to problem 1, the frequency feature learning branch adopts a novel Fourier-based feature extractor for efficient feature extraction. Moreover, the spectral feature learning branch introduces the band grouping method to explore the spectral reflectance characteristics of HSIs, which serves as a solution to problem 2. Last but not least, the spatial feature learning branch comprises a CEEM to enhance feature discrimination, which is tailored to the characteristics of HSIs and the patch-based HSI classification pattern. This branch materialized from our contemplation on problems 1 and 3. With the help of effective fusion and advanced loss function, the multidomain features from three branches make the proposed model exhibit outstanding performance, which is evaluated and demonstrated by extensive experiments.

In the future, we will investigate improved techniques for harnessing and fusing multidomain features, e.g., using multiple loss functions to evaluate different features similar to multitask learning. In addition, we will not be limited to using hyperspectral data for classification. Domain adaptation [56], [57] related to hyperspectral data and information fusion with other data such as LiDAR data [58], [59] are also of our interest.

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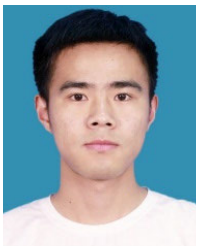
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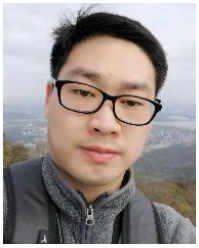
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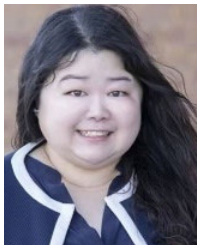
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